

Passive Drainage Priority Safety Floor

Executive Technology Brief

Domain: Hydraulic Passive Safety + Multi-Lumen Valve Mechanics

Technology Maturity: ENGINEERING_DEFINITION

Transfer Posture: Sponsored Validation

Problem: Obstruction causes 30-50% of shunt failures with no passive safety floor

Mechanism: Multi-lumen parallel catheter with differential conductance maintaining minimum drainage when primary path obstructs

Architecture: CSF shunt with passive safety floor: a secondary drainage path with lower conductance that maintains minimum drainage when the primary path obstructs. Differential pressure thresholds activate the floor path.

Governing Model: Parallel hydraulic conductance model. Two drainage paths in parallel: primary (high conductance) and floor (lower conductance). When primary obstructs, floor maintains minimum drainage.

Key Equations:

$Q = (\pi * r^4 * \Delta P) / (8 * \eta * L)$ [P-07 multi-lumen floor conductance – laminar flow per lumen]

$Q_{total} = Q_{primary} + Q_{floor}$

$G_{total} = G_{primary} + G_{floor}$ [parallel conductance] [P-07 parallel floor conductance – passive safety floor mechanism]

$Q_{floor_min} = G_{floor} * \Delta P_{max}$ [minimum drainage at worst-case pressure]

Kill Condition: Common-cause obstruction affects both primary and floor lumens equally (no differential immunity)

Current State: ENGINEERING_DEFINITION. No physical prototype. No clinical validation.

Disclosure: This dossier distinguishes established evidence from proposals and unresolved questions.